Exploring Fractions and Number with Virtual Manipulatives

Mathigon Polypad · Amplify

Years 7-8	~30 minutes	Mathematics · Number

NOTICE — AI-generated worksheet
This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: https://polypad.amplify.com/p
Curriculum links: AC9M7N06 · AC9M7N09 · AC9M8N04 · AC9M8N05

Learning intentions

- I can use virtual fraction bars to show equivalent fractions and compare their sizes.
- I can use the four operations with fractions, decimals and percentages to solve a practical problem.
- I can choose and justify an efficient strategy for solving a rational-number problem.

Getting started

Open Mathigon Polypad and select the Fraction Bars tool from the manipulatives menu. Drag bars onto the canvas to build the fractions in each task below, then switch to the algebra tiles tool where a later task asks you to.

Tasks

1. Build fraction bars for 1/2 and 2/4 and place them side by side. State whether they are equivalent and how you can tell from the bars.

2. Build bars for three given fraction pairs and record whether each pair is equivalent.

Fraction pair	Equivalent? (yes/no)

3. Use fraction bars to model adding two fractions with different denominators, such as 1/3 + 1/4. Show your working alongside a short description of what you did with the bars.

Working:

4. Convert one of your fractions to a decimal and to a percentage. Show all three forms.

5. A $60 jacket is reduced by 25%. Use Polypad tools of your choice to model and solve this problem, showing an efficient strategy.

Working:

6. Explain why building fractions physically as bars can make it easier to check whether an addition or comparison answer makes sense.

7. Switch to the algebra tiles tool and build a simple expression such as 2x + 3. Describe what each type of tile represents.

Extension (optional)

Use Polypad to model a shopping scenario involving a percentage discount followed by a percentage tax added afterwards. Explain which order (discount first or tax first) gives a lower final price, and why.